Alireza Shahi

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Research Profile

Mathematics Ph.D. researcher developing mathematically structured machine-learning methods for high-dimensional data. Research spans geometric and topological representations, statistical learning, deep learning on 3D structured data, and rigorous analysis of model generalization. Interested in theoretical and empirical questions in continual learning, representation dynamics, optimization, and learning in deep neural networks.

Education

University of Tennessee, Knoxville Expected 2028
Ph.D. Student, Mathematics and Statistics — GPA: 3.93/4.00

- Relevant coursework: Probability I–II, Advanced Probability I, Statistics I–II, Statistical Learning, Data Mining, Applied Time Series, Numerical Analysis I–II, Mathematical Methods for Deep Learning.

Technical Skills

Programming: Python, R, MATLAB, SQL, LaTeX
Machine Learning: PyTorch, scikit-learn, NumPy, pandas, SciPy, gradient boosting, random forests, neural networks, point transformers
Methods: representation learning, feature engineering, regression, classification, cross-validation, bootstrap methods, permutation testing, model selection, ablation studies, sensitivity analysis, dimensionality reduction, clustering, out-of-sample validation

Research Experience

Ph.D. Researcher – Mathematical Machine Learning 2023–Present
University of Tennessee

- Develop mathematically structured representations for high-dimensional scientific data, combining geometry, topology, statistical learning, and predictive modeling.
- Built reproducible Python pipelines for 4,057 protein–ligand complexes, including large-scale feature extraction, model training, five-fold cross-validation, independent holdout evaluation, bootstrap uncertainty analysis, permutation tests, and representation ablations.
- Developed **Element Interaction Manifolds (EIM)** and related geometric representations for protein–ligand recognition and binding-affinity prediction, emphasizing interpretable structure and robust out-of-sample validation.
- Developed **Chromatic-TDA** representations based on persistent homology and inclusion-induced topological features; designed matched ordinary-PH, dimensionality, metadata, and search-width controls to isolate predictive contributions of representation structure.
- Investigated trade-offs among representation dimension, predictive accuracy, robustness, and generalization using controlled model comparisons.

Researcher – Deep Learning for 3D Structured Data 2026
Multisensor Fusion for 3D Facial Landmark Detection

- Developed and evaluated a Point Transformer pipeline for 3D facial landmark prediction using irregular surface meshes.
- Integrated geometry, RGB texture, multi-scale curvature, bilateral symmetry, graph neighborhoods, and attention-based decoding into a unified deep-learning representation.
- Contributed to preprocessing, PyTorch model evaluation, visualization, interpretation, and experimental analysis in a multidisciplinary research team.

Selected Research and Publications

- **Element Interaction Manifolds for Comparative Analysis of Geometric Representations in Protein–Ligand Recognition.** First-author ChemRxiv preprint, 2026.
- **Chromatic Topological Data Analysis for Protein–Ligand Binding-Affinity Prediction.** First-author manuscript in preparation; develops inclusion-based topological representations and controlled predictive comparisons.
- **Multisensor Fusion for 3D Facial Landmark Detection in Unilateral Cleft Lip.** IEEE MFI conference paper, 2026; Point Transformer learning on 3D facial meshes.
- **SIAM MDS26:** "Differential Geometric Manifold Representations for Protein–Ligand Binding Affinity Prediction," Salt Lake City, 2026.